

Weekly Report – Week 4 Group-1

Project: *Time-Series Few-Shot Anomaly Detection for HVAC Systems*

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Existing Methods (Findings) By A.Alim

Method	Limitation
AE / LSTM-AE	Works only on a single domain
Residual-based	Poor generalization
Transfer Learning	Requires labeled anomalies
Domain Adaptation	Costly or inaccurate
One-Class SVM	Needs large normal dataset
Few-Shot Learning	Not domain adaptive

Implementation (Week 4) By Farea Mahdea

Baseline: Unsupervised LSTM Autoencoder

- Input: 17 sensor features, window size = 10 timesteps
- Encoder–Decoder with 2–3 LSTM layers
- Loss: MSE | Optimizer: Adam
- Output: Reconstruction error → Threshold-based anomaly detection

Data Preprocessing (MZVAV-2 Dataset)

Samples: 18,720 real + 37,440 simulated

- Features: 17 (temperature, pressure, valve signals, fan speed, etc.)
Steps: Clean missing values → Normalize (0–1) → Create time windows (10 steps) → Train/Val/Test split

Next Step: Trying few-shot fine-tuning on the target domain to improve adaptation and reduce retraining effort. By Anik Barua

